

CERTIFIED GROUNDWATER QUALITY & HEAVY METAL FIELD REPORT

Smart India Hackathon SIH25067 | Tamil Nadu Water Supply & Drainage Board (TWAD)

Borewell / Location:	Kadaladi Monitoring Station #4 (Field Demonstration)	Date & Time:	17-09-2026 17:26 IST
GPS Coordinates:	9.22200 N, 78.49600 E	Hydrological Season:	Post-Monsoon
Evaluation Mode:	Mode A (Analytical Laboratory)	Inspection Authority:	CIT ECE Water Intelligence Group

HPI SCORE: 72.4	EVALUATION (HEI): 4.82	CRITICAL HAZARD - NON-POTABLE
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1. Hydrochemical Parameters & Standard Compliance (BIS IS 10500:2012)

Parameter	Concentration	Permissible (SI)	Status / Excess
pH	7.3	6.5 - 8.5	Compliant
TDS	1954.0	500 mg/L	Elevated
EC	2877.0	750 uS/cm	Elevated
Cadmium (Cd)	0.0038 mg/L	0.0030 mg/L	Exceeds Permissible (+26.7%)
Iron (Fe)	0.5800 mg/L	0.3000 mg/L	Exceeds Permissible (+93.3%)
Manganese (Mn)	0.3000 mg/L	0.1000 mg/L	Exceeds Permissible (+200.0%)

2. USEPA Human Health Risk & Toxicological Assessment (RAGS Model)

Child Non-Carcinogenic Hazard Index (HI): 3.24 [Severe Chronic Danger] | Adult Hazard Index (HI): 1.42 [Health Advisory]

Primary Toxicological Hazard Driver: Cadmium (Cd) (Targeting renal and neurological pathways)

3. Actionable Engineering Remediation & Community Treatment Plan

Primary Verdict: Lime Neutralization + Reverse Osmosis Desalination Required

Estimated Treatment Cost: Rs. 38.50 / kL (1000 Liters)

Recommended Engineering Process Flow:

- Stage 1 - Coagulation/Flocculation: Lime ($\text{Ca}(\text{OH})_2$) Dosing — Precipitate Cadmium and Iron hydroxides (Est: Rs. 8.00/kL)
- Stage 2 - Granular Filtration: Multi-Media Sand & Carbon Filter — Trap particulate flocs and organic matter (Est: Rs. 4.50/kL)
- Stage 3 - Membrane Desalination: Polyamide Reverse Osmosis (RO) — Reduce salinity (TDS) and residual trace ions (Est: Rs. 26.00/kL)

4. Regional Spatial Location Verification

